

other examples of entrainment. The observed effects also differ essentially from the resonance drift induced by periodic forcing of an active medium¹² which occurs without synchronization and leads to displacement of the spiral-wave core as a whole, without the pronounced deformations of the tip trajectory²² characteristic of the described phenomenon. This new way to deform and translate a spiral wave by controlling excitability with even weak external forces could ultimately prove relevant to our understanding of pattern formation in the early stages of morphogenesis², or to the origin and possible control of abnormalities of heart rhythm¹. □

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Self-assembling organic nanotubes based on a cyclic peptide architecture

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HOLLOW tubular structures of molecular dimensions may offer a variety of applications in chemistry, biochemistry and materials science. Concentric carbon nanotubes^{1,2} have attracted a great deal of attention, while the three-dimensional tubular pore structures of molecular sieves have long been exploited industrially^{3–8}. Nano-scale tubes based on organic materials have also been reported previously^{9–13}. Here we report the design, synthesis and characterization of a new class of organic nanotubes based on rationally designed cyclic polypeptides. When protonated, these compounds crystallize into tubular structures hundreds of nanometres long, with internal diameters of 7–8 Å. Support for the proposed tubular structures is provided by electron microscopy, electron diffraction, Fourier-transform infrared spectroscopy and molecular modelling. These tubes are open-ended, with uniform shape and internal diameter. We anticipate that they may have possible applications in inclusion chemistry, catalysis, molecular electronics and molecular separation technology.

The proton-triggered self-assembly process described here is a highly convergent approach in which numerous ring-shaped peptide subunits interact through an extensive network of hydrogen bonds to form nanotube structures (Fig. 1). The present design features an eight-residue cyclic peptide with the following sequence: *cyclo*[-(D-Ala-Glu-D-Ala-Gln)₂]. In designing the subunit, we postulated that cyclic peptides with an even number of alternating D- and L-amino acids can adopt (or sample) a low-energy ring-shaped flat conformation in which all backbone amide functionalities lie approximately perpendicular to the plane of the structure. In this conformation, subunits can stack in an antiparallel fashion and participate in backbone-backbone intermolecular hydrogen bonding to produce a contiguous β -sheet structure. Moreover, because of the alternating D- and L-amino acid sequence, peptide side chains must necessarily lie on the outside of the ensemble thereby creating the desired hollow tubular core structure. Here, we chose to exploit the ionization state of the glutamic acid side chain functionality as the trigger

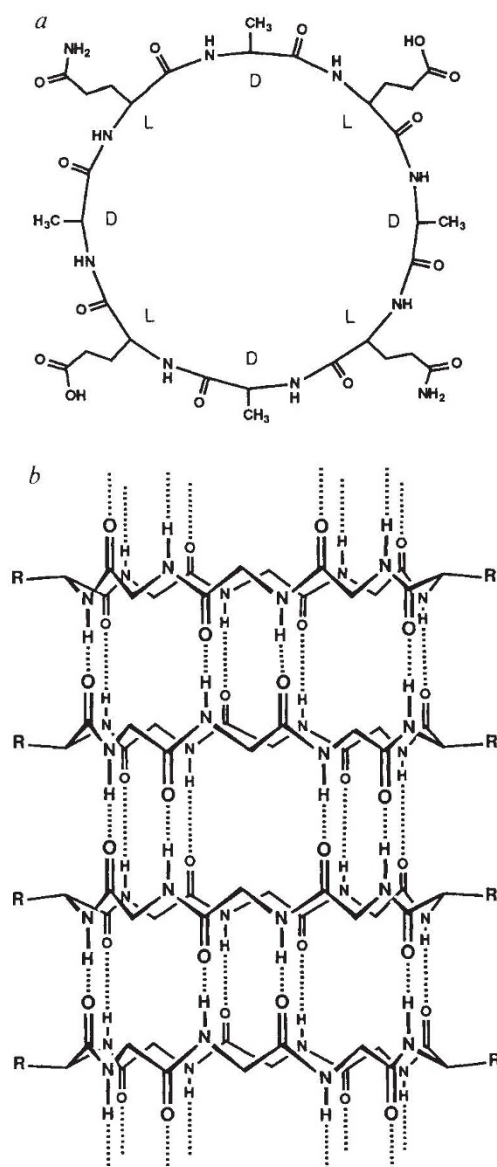
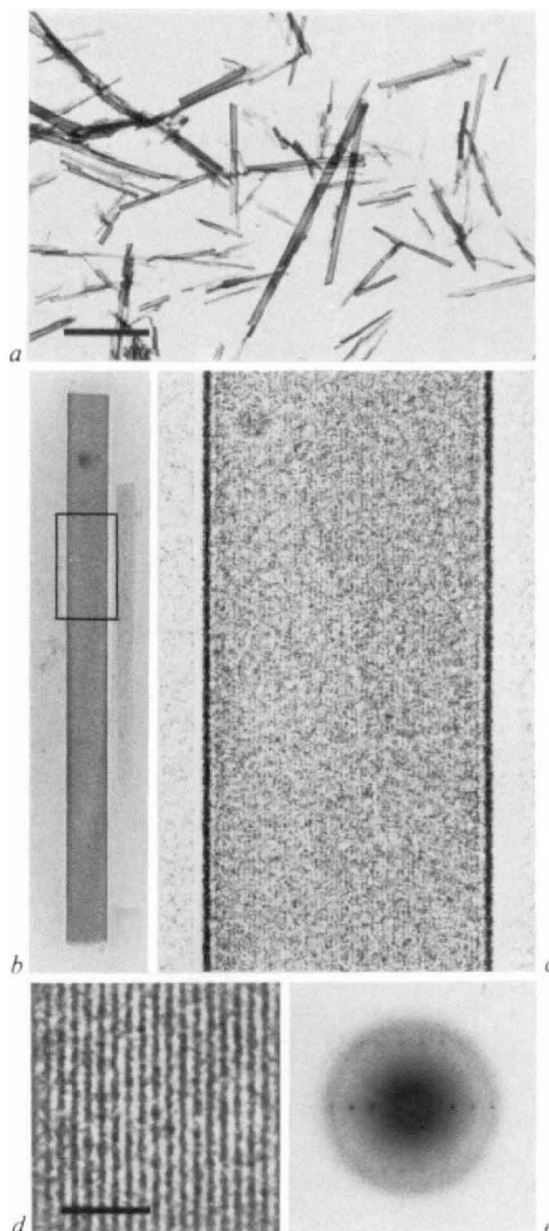


FIG. 1 a, A two-dimensional representation of the chemical structure of the peptide subunit (D or L refers to the amino acid chirality). b, Peptide subunits are shown in a self-assembled tubular configuration emphasizing the antiparallel stacking and the extensive network of intermolecular hydrogen-bonding interactions (for clarity only backbone structure is represented).

FIG. 2 Electron microscopy and electron diffraction of a nanotube particles. *a*, Low-magnification image of nanotube suspension adsorbed on carbon support film (scale bar, 1 μm). *b*, Low-dose image of a frozen hydrated single nanotube particle. The particle measures $\sim 86 \times 1180$ nm. The boxed region is enlarged in *c*, where the longitudinal striations can be seen most easily by viewing from the bottom at a glancing angle. Radiation damage and mechanical instabilities in the cold stage limit the attainable resolution in these images to ~ 10 Å. *d*, Image enhancement (see below) clearly reveals the 14.9-Å longitudinal striations. Presumably, this is due to the side-by-side packing of nanotubes in the particle (scale bar, 10 nm). *e*, Electron diffraction pattern recorded from a single nanotube particle showing orders of a 14.92-Å meridional spacing and a 4.73-Å axial spacing. Axially the pattern extends weakly to the third order (1.57 Å, data not shown) demonstrating that the nanotube particles are highly ordered and crystalline.

METHODS. The peptide was synthesized by the solid phase method²⁴ and characterized by $^1\text{H-NMR}$ spectroscopy, elemental analysis and ion-spray mass spectrometry. Although a variety of conditions may be used in the self-assembly of organic nanotubes, the following procedure has provided the most consistent results. A suspension (~ 25 mg ml^{-1}) of peptide subunit was clarified by the addition of 2.5 equivalents of NaOH. The resulting peptide solution was centrifuged to remove traces of solid matter and then acidified by the addition of one-third its volume of 1% trifluoroacetic acid in acetonitrile. Nanotube particles gradually form as a white suspension over a period of hours. Nanotubes may then be collected by centrifugation and washed repeatedly with distilled water to remove excess acids and salts. For electron microscopy and diffraction studies, a suspension of nanotube particles was exposed to ultrasound briefly and small drops applied to glow-discharged carbon support films on electron microscope grids. Excess liquid was removed by blotting, and the grids frozen in liquid ethane slush^{25,26}. Grids were mounted in a Gatan cold stage and examined in a Philips CM12 electron microscope operating at 120 kV. The specimen temperature was -175°C during examination and imaging. Images were recorded at $\times 35,000$ using strict low-dose conditions at various defocus levels. For image analysis, micrographs were converted to optical-density arrays using a Perkin-Elmer scanning microdensitometer with spot and step sizes equal to 2.86 Å at the specimen. Using the SUPRIM program package²⁷, a number of small areas from a single TEM image were rotationally and translationally aligned and then averaged.



mechanism for the initiation of the self-assembly process in aqueous solutions. It was expected that at alkaline pH, the large repulsive intermolecular electrostatic interactions between the negatively charged carboxylate side chains would disfavour ring stacking and at the same time promote the dissolution of the peptide subunit in aqueous media. We postulated that, on protonation of the carboxylate groups, not only would the unfavourable intermolecular electrostatic interactions vanish, but attractive side-chain/side-chain hydrogen bonding should occur. Furthermore, at acidic pH the peptide subunit is expected to be less soluble in aqueous media which was thought to contribute further to an ordered phase transition toward self-assembled nanotube particles. We therefore concluded that as intersubunit hydrogen bonding provides the principal driving force in the assembly process, and considering that only in the stacked configuration can the peptide subunits enjoy the maximum possible number of hydrogen-bonding contacts, protonation of the carboxylate groups should strongly bias the system towards self-assembly.

Controlled acidification of alkaline peptide solutions triggers the spontaneous self-assembly of peptide subunits into rod-shaped crystals (Fig. 2*a* and *b*). Transmission electron micros-

copy (TEM) indicates that each crystal is an organized bundle of hundreds of tightly packed nanotubes. Low-dose cryomicroscopy of the crystals reveals longitudinal striations with spacing of 14.9 Å along the long axis of the crystal as expected for the centre to centre spacing for closely packed nanotubes (Fig. 2*c* and *d*). Electron diffraction patterns show 4.73-Å axial periodicity (Fig. 2*e*), suggesting that each nanotube is made up of stacked rings with intersubunit distances corresponding to an ideal antiparallel β -sheet structure^{14,15} (Fig. 1*b*). Given the thin rod shape of the crystals—on average 10–30 μm long and 100–500 nm wide—it is reasonable to assume that crystals would lie on the supporting electron microscope carbon film along the long (*a*) axis and either the *a*–*b* or *a*–*c* faces. All micrographs and diffraction patterns, however, show similar diffraction intensities and nearly identical lattices. This suggests that the *b*- and *c*-axes are identical in size within experimental error. Furthermore, given the highly symmetric nature of the octapeptide, it seems reasonable that the similarity of the views reflects the symmetry of the molecule. Electron diffraction patterns showed a unit cell having a meridional spacing of 14.92 ± 0.08 Å and an axial spacing of 4.73 ± 0.02 Å with an angle of $99.2 \pm 0.5^\circ$. The diffraction patterns did not show any symmetry other than the centre of

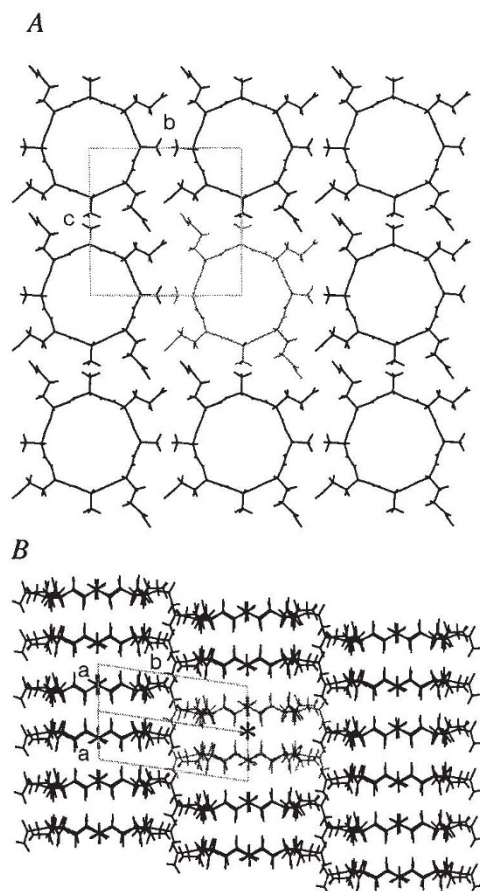


FIG. 3 Three-dimensional model of the self-assembled organic nanotubes (see text for the details of model building). View of the crystal packing along the *a*-axis (A) and the *c*-axis (B) are shown. The unit cell is indicated by the solid lines and the local 2-fold axis by the asterisk. The unit cell has dimensions $a=9.5$ Å, $b=c=15.1$ Å, and $\alpha=90^\circ$, $\beta=\gamma=99^\circ$. The tube axis is along *a*.

symmetry due to Friedel's law. The patterns extend to 1.5 Å and occasionally to 1.2 Å indicating that the crystals are highly ordered. Considering the above observations, it follows that the lattice is trigonal with $a=4.73$ Å, $b=c=15.1$ Å, $\alpha=90^\circ$ and $\beta=\gamma=99^\circ$.

Although many conformers are possible using reasonable peptide backbone ϕ , ψ angles, only one fits the box defined by the unit cell—the disk-shaped flat conformation where all side chains extend outwards in the plane of the disk. Other conformations are puckered, where the irregular presentation of side chains and backbone conformation would prevent the packing of peptide subunits to within 4.7 Å due to prohibitive van der Waals contacts. The only way to pack two peptides to within 4.7 Å is to have the backbone carbonyl functionalities hydrogen-bond intermolecularly to the nitrogen amide moieties of the opposite chain and position the side chains in the plane of the peptide backbone in order to avoid unfavourable side chain/side-chain steric interactions. Furthermore, owing to the use of alternating D- and L-amino acid residues, the most favourable intermolecular hydrogen bonds form when the disks are stacked on top of one another in an antiparallel fashion giving rise to local 2-fold symmetry (Fig. 3). The model built in this way, using the program XtalView¹⁶, has unit cells that might at first glance seem twice as large ($9.5 \times 15.1 \times 15.1$), which would require a spacing of 9.4 Å in the diffraction pattern. But when diffraction patterns are calculated, every other level of *h* cancels out due to the pseudo-symmetry of the dimer. That is, the ring flipped over is

nearly equivalent to the unflipped ring (Figs 1 and 3). Several other packing arrangements were tried but none were successful in forming a three-dimensional lattice without spatial overlaps and simultaneously fitting the spacing indicated by the electron diffraction patterns. Cells where the local 2-fold symmetry axis of the dimer is crystallographic were also considered but then the lattice could either not be modelled or the diffraction pattern had to have a mirror plane which is not observed. One other possibility is to consider that the observed diffraction pattern shows a diagonal of the lattice instead of a principle axis. This

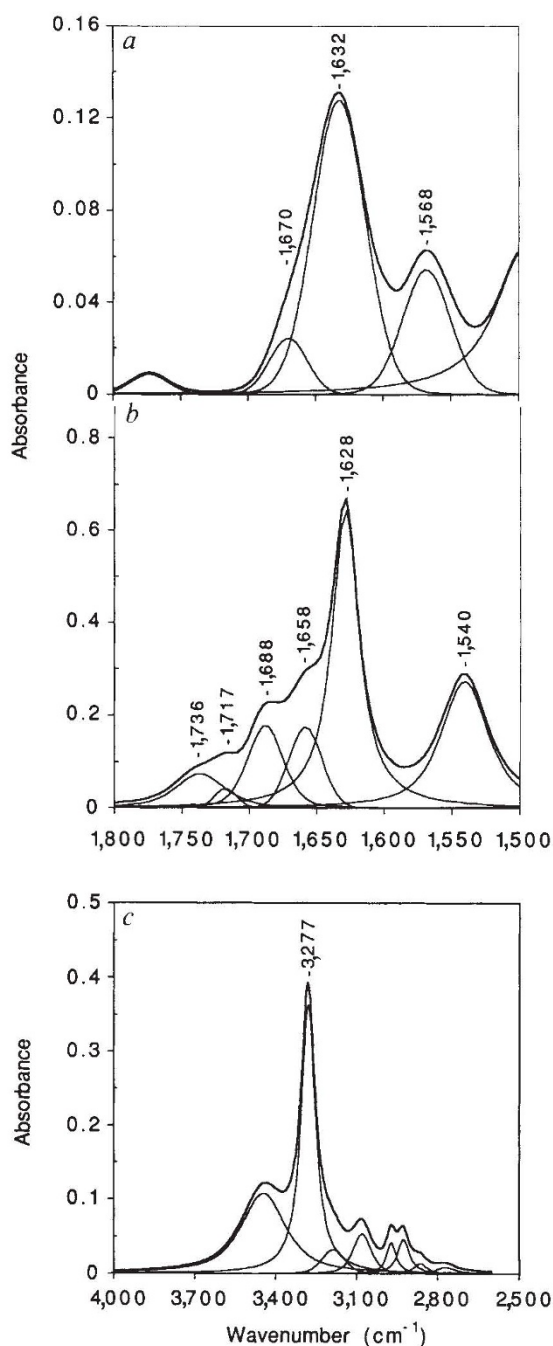


FIG. 4 Infrared spectra (at 8-cm^{-1} resolution) of amide-I region of *a*, monomeric peptide subunit in D_2O (4×10^{-3} M, $\text{pD}=10$, examined by attenuated total reflectance and *b*, nanotube particles (KBr pellet). *c*, N-H stretch region of the FT-IR spectrum of nanotube particles (KBr pellet). Component peaks were obtained by deconvolution of the original spectrum with single component, mixed lorentzian and gaussian functions using an iterative, linear least-squares algorithm (FIT, Galactic Industries).

can be easily dismissed because it requires a substantially smaller unit cell. It should be pointed out that the volume of the unit cell in our proposed model is just large enough to contain the octapeptide dimer and anything smaller can not physically hold the entire mass of the octapeptide. Therefore, the observed electron diffraction pattern must be due to the principle axis. Finally the crystalline model was checked for bad contacts using the program XPLOR¹⁷ (molecular dynamics calculations on the three-dimensional crystal lattice). No bad contacts were found and the overall energy of the model was low. Most importantly, when the three-dimensional model was used to calculate structure factors, the patterns produced compared very favourably with the electron diffraction patterns thus strongly supporting the proposed model.

Involvement of intermolecular hydrogen bonding networks in tube assembly is also supported by Fourier-transform infrared (FT-IR) spectroscopic analysis¹⁸ (Fig. 4). In alkaline solutions, the monomeric peptide subunit displays an amide-I band at 1,632 cm⁻¹ consistent with the cyclic structure of the backbone. Moreover, the bands at 1,568 and 1,670 cm⁻¹ are typical solvent-exposed carbonyl stretching frequencies for glutamate and glutamine side-chain functionalities, respectively. On self-assembly, however, organic nanotubes display characteristic features of a hydrogen-bonded β -sheet structure. Not only is the appearance of amide-I bands at 1,628 cm⁻¹, and 1,688 cm⁻¹, and the amide-II band at 1,540 cm⁻¹ consistent with the expected backbone structure, but the observed N-H stretching frequency at 3,277 cm⁻¹ also strongly supports formation of a tight network of backbone-backbone hydrogen bonding. Given the tight backbone-backbone inter-subunit interactions, as determined by the above analysis of the electron diffraction patterns, it is possible to use Krimm's correlation to estimate the intermolecular hydrogen-bonding distance from the observed frequency of the N-H stretching vibration¹⁸. The observed frequency of N-H stretching mode approximately correlates to an average inter-subunit N-O distance of 2.85 Å, or an inter-subunit distance of 4.71 Å. This value is consistent with the value of 4.73 Å obtained from electron diffraction patterns. It should be pointed out that the IR spectrum of the nanotubes also closely resemble the ones reported for gramicidin A—a naturally occurring linear peptide composed of hydrophobic amino acids with alternating D- and L-configuration which is known to form dimeric β -helical transmembrane ion channel structures^{19,20}. Gramicidin A ($\uparrow\downarrow\beta^{6,4}$, that is, an antiparallel β -helix with a 6.4 residue rise per turn) has amide-I bands at 1,630, 1,685 cm⁻¹, an amide-II band at 1,539 cm⁻¹, and an N-H stretching frequency at 3,285 cm⁻¹ (ref. 21).

The general strategy described here should allow the design and synthesis of a wide range of tubular structures with specified internal diameters and surface characteristics, and having wide range of applications^{22,23}. For example, such tubular materials could be designed to mimic biological channels and pore structures, used to study physical and chemical properties of confined molecules, control growth and properties of inorganic and metallic clusters, or design novel optical and electronic devices. □

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The contribution of explosive volcanism to global atmospheric sulphur dioxide concentrations

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SULPHUR dioxide from volcanic eruptions may have a significant effect on the Earth's climate and atmospheric chemistry, and it is therefore important to quantify outgassing rates for all types of volcanic activity. Non-explosive volcanoes (for example, Mount Etna) outgas at relatively constant rates, providing an annual flux of about 9 million tons (Mt) SO₂ (ref. 1). By contrast, the outgassing from volcanoes prone to explosive eruptions (such as Mount Pinatubo) is sporadic and much more difficult to quantify. The total annual volcanic SO₂ flux is therefore poorly constrained, with ground-based estimates^{1–3} ranging from 1.5 to 50 Mt—up to one-quarter of the estimated current anthropogenic contribution. The Total Ozone Mapping Spectrometer aboard the NASA satellite Nimbus 7 recorded SO₂ emissions from explosive eruptions from November 1978 to May 1993. We use these data to show that the annual flux from explosive volcanism is of the order of 4 Mt SO₂, less than half of the non-explosive output. Thus it seems that the total volcanic emission of SO₂ to the Earth's atmosphere is about 13 Mt yr⁻¹, which is only 5–10% of the current anthropogenic flux.

The recent eruption of Mount Pinatubo has rejuvenated the question of volcanic effects on global climate. The climatic impact of volcanic activity is potentially large because, in addition to the relatively consistent degassing of non-explosive volcanoes into the Earth's atmosphere, the infrequent bursts of explosive volcanoes such as Pinatubo have the ability to inject sulphur dioxide directly into the stratosphere. However, constraints on the SO₂ production from global volcanism have been elusive. Previous estimates of the annual sulphur dioxide flux due to volcanic activity, using ground-based methods, vary by more than a factor of 30 (Table 1)^{1–8}. These estimates range from insignificant amounts to >25% of the present annual anthropogenic flux of ~190 million tons (Mt)⁹. Previous efforts to partition global SO₂ production into explosive and non-